No. 7-Dars

DATE 08.11.2026 Feature Engineering

Feature bilan ishlash, yangi featurelar yaratish bir turdan ikkinchi turga transform qilish va muhim featurelarni tanlash bosqichi.

Uning maqsadi model performanceni yaxshilash va Model generalizationni oshirish.

Model performance → modelning natijasi.

Regression → MAE, MSE, RMSE, R^2

classification → Accuracy, Precision, Recall, F_1-score

Generalization → modelning o'zi o'rganmagan yangi ma'lumotlarda ham yaxshi natija ko'rsata olish qobilyati.

Feauture engineering techniques.

1. Handling missing values
2. Encoding
3. Scaling
4. Feature creation
5. Feature selection.

Feature creation → mavjud featurelardan yangi feature yaratish.

Feature creation 3 ta yo'l bilan amalga oshiri-ladi.

1. Matematik
2. Domain knowledge
3. Time based

DATE ( )

Feature selection → dataset ichida muhim featurelarni tanlash va foydasiz futurelarni olib tashlash usullari

1- correlation → featurelar va target orasidagi bog'liqlikni o'rganadi.

2- Feature importance → Random forest kabi model orqali featurelarning ahamiyatini baxo

3- Recursive Feature Elimination (RFE)
Modelni qayta qayta train qilib, kamroq muhim featurelarni olib tashlash

4- select best → statistika orqali eng yaxshi k ta featurelarni tanlash